Part A

Prompt: Using the csv file attached, form daily log returns for the eight stocks (GM, Ford, UTX, CAT, Merck, Pfizer, IBM, Microsoft), and fit a two-factor model with factanal(..., factors = 2, rotation = "none"). Report the factor loadings and the uniquenesses.

AI Code:

```
# Two-factor model on daily log returns of eight stocks
# Data: Stock_FX_Bond.csv

data <- read.csv("Stock_FX_Bond.csv", header = TRUE)

# Select adjusted-close columns for the eight stocks
tickers <- c("GM_AC", "F_AC", "UTX_AC", "CAT_AC", "MRK_AC", "PFE_AC", "IBM_AC",
"MSFT_AC")
prices <- data[, tickers]

# Compute daily log returns: r_t = log(P_t) - log(P_{t-1})
log_returns <- as.data.frame(apply(log(prices), 2, diff))
colnames(log_returns) <- c("GM", "Ford", "UTX", "CAT", "Merck", "Pfizer", "IBM", "Microsoft")

cat("Number of return observations:", nrow(log_returns), "\n\n")

# Fit two-factor model, no rotation
fa_fit <- factanal(log_returns, factors = 2, rotation = "none")

print(fa_fit, digits = 3, cutoff = 0)
```

Outputs:

```
Number of return observations: 4962

Call:
factanal(x = log_returns, factors = 2, rotation = "none")

Uniquenesses:
   GM    Ford    UTX    CAT   Merck  Pfizer
 0.399  0.423  0.718  0.714  0.519  0.410
  IBM Microsoft
 0.760  0.749

Loadings:
        Factor1 Factor2
```

GM	0.693	-0.348
Ford	0.692	-0.313
UTX	0.531	-0.028
CAT	0.529	-0.079
Merck	0.551	0.421
Pfizer	0.574	0.511
IBM	0.490	0.004
Microsoft	0.499	0.044

	Factor1	Factor2
SS loadings	2.643	0.666
Proportion Var	0.330	0.083
Cumulative Var	0.330	0.414

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 564.66 on 13 degrees of freedom.
The p-value is 2.6e-112

Part B

HW5 Part B

1. Ford: $\lambda_1 = 0.692 \quad \lambda_2 = -0.313$

$h^2_{Ford}: \lambda_1^2 + \lambda_2^2 = (0.692)^2 + (-0.313)^2 = 0.576833$

$\psi_{Ford} = 1 - h^2 = 1 - 0.576833 = 0.423167$

GM: $\lambda_1 = 0.693 \quad \lambda_2 = -0.348$

$h^2_{GM} = (0.693)^2 + (-0.348)^2 = 0.601353$

$\psi_{GM} = 1 - 0.601353 = 0.398647$

Both ψ_{Ford} and ψ_{GM} are the same as the uniqeness reported by factanal.

2. $\hat{\rho}(F, IBM) = \lambda_{F_1} \cdot \lambda_{IBM_1} + \lambda_{F_2} \cdot \lambda_{IBM_2} = (0.692)(0.490) + (-0.313)(0.004)$

$= 0.337828$

$\hat{\rho}(F, IBM) \approx 0.338$ and matches the off-diagonal of $\Lambda\Lambda^T + \Psi$

Part C

The small p-value is not evidence that two factors are enough, but rather the opposite. While it is true that the p-value is “overwhelmingly significant,” it is strongly rejecting the null hypothesis that two factors are sufficient. This means that two factors are not enough. Another error is that communality is not the same as uniqueness, they cannot be used interchangeably. The uniqueness is 0.423, as seen by the R code outputs and the hand calculation from part B.

Part D

1. Factor 1 represents the general market factor that loads positively and fairly uniformly on all eight stocks. It captures broad macroeconomic movements like GDP changes, market-wide sentiment, or interest rate fluctuations. Factor 2 is an industry-specific or sector-rotation factor, it is strongly negative for GM and Ford, strongly positive for Merck and Pfizer, and close to zero for UTX, CAT, IBM, and Microsoft.
2. A stock's uniqueness is the share of its return variance that the common factors don't explain, ie. a risk specific to that company. If you hold many stocks, their idiosyncratic (unique) risks tend to cancel, leaving mostly factor risk. High uniqueness means most of the stock's day-to-day variances is company specific, but two-factor model explains relatively little of what moves that particular stock.
3. The model assumes that control for the common factors, the remaining stock-specific shocks are uncorrelated with each other. However, this is not necessarily true in the real-world. For example, Ford and GM can still be correlated outside the two factors because they are both in the automobile industry, which can be volatile to gas prices, supply-chain shocks, and government regulations.